

JavaScript

Lab Assignment

JavaScript Introduction

Task:

- Create a simple HTML page and add a <script>tag within the page.
- Write JavaScript code to display an alert box with the message "Welcome toJavaScript!" when the page loads.

Ans :

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript Alert</title>
</head>
<body>

<h1>My Web Page</h1>

<script>
  alert("Welcome to JavaScript!");
</script>

</body>
</html>
```

Variables and Data Types

Task:

- Write a JavaScript program to declare variables for different data types (string,number, boolean, null, and undefined).
- Log the values of the variables and their types to the console using console.log().

Ans :

```
let name = "Rohan";

let age = 20;
```

```

let isStudent = true;

let emptyValue = null;

let notAssigned;

console.log("Name:", name, "| Type:", typeof name);
console.log("Age:", age, "| Type:", typeof age);
console.log("Is Student:", isStudent, "| Type:", typeof isStudent);
console.log("Empty Value:", emptyValue, "| Type:", typeof emptyValue);
console.log("Not Assigned:", notAssigned, "| Type:", typeof notAssigned);

```

JavaScript Operators

Task:

- Create a JavaScript program to perform the following:
- Add, subtract, multiply, and divide two numbers using arithmetic operators.
- Use comparison operators to check if two numbers are equal and if one number is greater than the other.
- Use logical operators to check if both conditions (e.g., $a > 10$ and $b < 5$) are true.

Ans :

```

let a = 15;
let b = 4;

console.log("Addition:", a + b);
console.log("Subtraction:", a - b);
console.log("Multiplication:", a * b);
console.log("Division:", a / b);

console.log("Are a and b equal?", a == b);
console.log("Is a greater than b?", a > b);

console.log("Is (a > 10 AND b < 5)?", (a > 10 && b < 5));

```

Task1:

- Write a JavaScript program using a for loop to print numbers from 1 to 10.

Ans :

```

for (let i = 1; i <= 10; i++) {
    console.log(i);
}

```

Task2:

- Create a JavaScript program that uses a whileloop to sum all even numbers between 1 and 20.

Ans :

```
let i = 2;
let sum = 0;

while (i <= 20) {
    sum += i;
    i += 2; // move to next even number
}

console.log("Sum of even numbers:", sum);
```

Task3:

- Write a do-whileloop that continues to ask the user for input until they enter a number greater than 10.

Ans :

```
let num;

do {
    num = parseInt(prompt("Enter a number greater than 10:"));
} while (num <= 10);

console.log("You entered:", num);
```

Functions

Task1:

- Write a function greetUserthat accepts a user's name as a parameter and displaysa greeting message (e.g., "Hello, John!").

Ans :

```
function greetUser(name) {
    console.log("Hello, " + name + "!");
}

greetUser("John");
```

Task2:

- Create a JavaScript function calculateSum that takes two numbers as parameters, adds them, and returns the result.

Ans :

```
function calculateSum(a, b) {  
    return a + b;  
}  
  
let result = calculateSum(5, 10);  
console.log("Sum:", result);
```

Arrays

Task1:

- Declare an array of fruits (["apple", "banana", "cherry"]). Use JavaScript to:
 - Add a fruit to the end of the array.
 - Remove the first fruit from the array.
 - Log the modified array to the console.

Ans :

```
let fruits = ["apple", "banana", "cherry"];  
  
fruits.push("orange");  
  
fruits.shift();  
  
console.log("Updated fruits:", fruits);
```

Task2:

- Write a program to find the sum of all elements in an array of numbers.

Ans :

```
let numbers = [10, 20, 30, 40];  
let sum = 0;  
  
for (let i = 0; i < numbers.length; i++) {  
    sum += numbers[i];  
}
```

```
console.log("Sum of array:", sum);
```

Objects

Task:

- Create a JavaScript object car with properties brand, model, and year. Use JavaScript to:
 - Access and print the car's brand and model.
 - Update the year property.
 - Add a new property color to the car object.

Ans :

```
let numbers = [10, 20, 30, 40];
let sum = 0;

for (let i = 0; i < numbers.length; i++) {
    sum += numbers[i];
}

console.log("Sum of array:", sum);
```

```
let car = {
    brand: "Toyota",
    model: "Corolla",
    year: 2020
};

console.log("Brand:", car.brand);
console.log("Model:", car.model);

car.year = 2023;

car.color = "Black";

console.log("Updated car object:", car);
```

JavaScript Events

Task:

- Create a simple webpage with a button that, when clicked, displays an alert saying "Button clicked!" using JavaScript event listeners.

Ans :

```
<!DOCTYPE html>
<html>
<head>
  <title>Button Click Event</title>
</head>
<body>

<h2>Click the Button</h2>
<button id="myBtn">Click Me</button>

<script>
  let button = document.getElementById("myBtn");

  button.addEventListener("click", function() {
    alert("Button clicked!");
  });
</script>

</body>
</html>
```

DOM Manipulation

Ans :

```
<!DOCTYPE html>
<html>
<head>
  <title>DOM Manipulation</title>
</head>
<body>

<p id="text">Hello, World!</p>

<script>
  let para = document.getElementById("text");

  para.textContent = "JavaScript is fun!";

  para.style.color = "blue";
</script>

</body>
</html>
```

JavaScript Timing Events (setTimeout, setInterval)

Task1:

- Write a program that changes the background color of a webpage after 5 seconds using setTimeout().

Ans :

```
<!DOCTYPE html>
<html>
<head>
  <title>Change Background</title>
</head>
<body>

<h2>Wait for 5 seconds...</h2>

<script>
  setTimeout(function() {
    document.body.style.backgroundColor = "lightblue";
  }, 5000); // 5000 ms = 5 seconds
</script>

</body>
</html>
```

Task2:

- Create a digital clock that updates every second using setInterval().

Ans :

```
<!DOCTYPE html>
<html>
<head>
  <title>Digital Clock</title>
</head>
<body>

<h2 id="clock"></h2>

<script>
  function updateClock() {
    let now = new Date();
    let time = now.toLocaleTimeString();
    document.getElementById("clock").textContent = time;
  }
</script>
```

```
    setInterval(updateClock, 1000);  
</script>  
  
</body>  
</html>
```

JavaScript Error Handling

Task:

- Write a JavaScript program that attempts to divide a number by zero. Use trycatch to handle the error and display an appropriate error message.

Ans :

```
<!DOCTYPE html>  
  
<html>  
<head>  
  <title>Try Catch Example</title>  
</head>  
<body>  
  
<h2>Check Console for Output</h2>  
  
<script>  
  let a = 10;  
  let b = 0;  
  
  try {  
    if (b === 0) {  
      throw new Error("Cannot divide by zero!");  
    }  
  
    let result = a / b;  
    console.log("Result:", result);  
  
  } catch (error) {  
    console.log("Error:", error.message);  
  }  
</script>  
  
</body>  
</html>
```